day 09 | 250917 W

heretofore

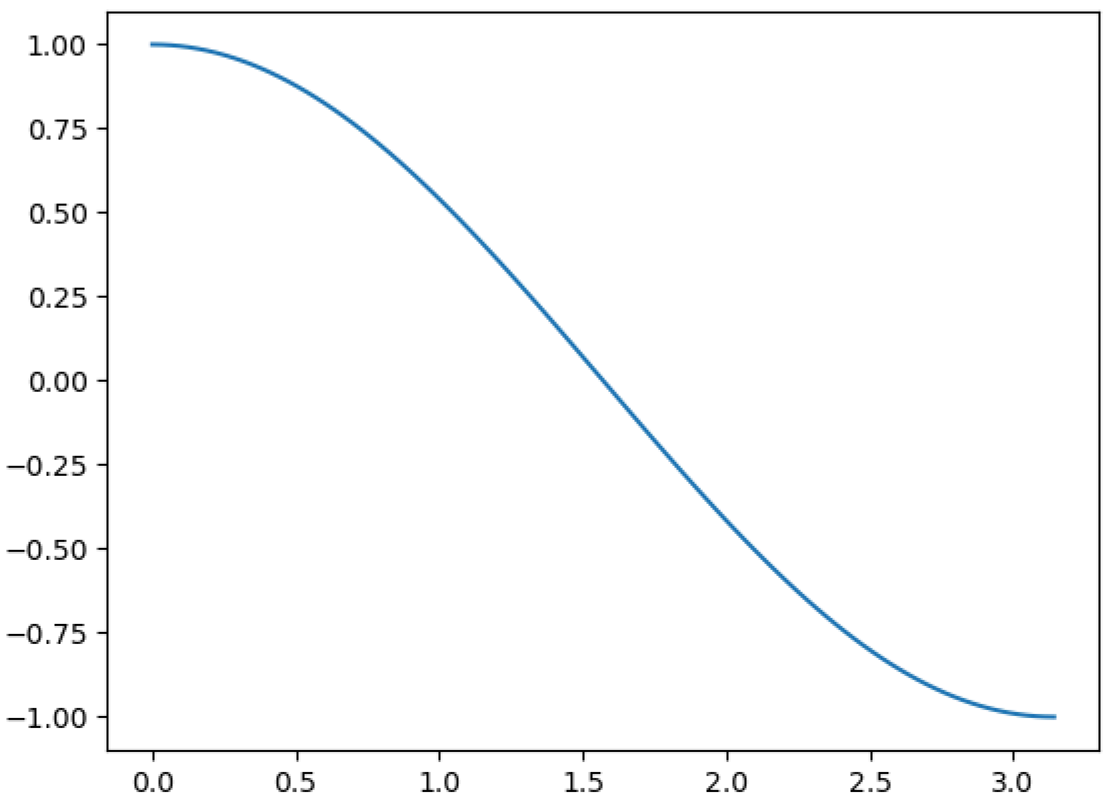
We first worked on some homework questions that had some difficult things within them.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [12]:
```
x = np.linspace(0,np.pi/2, 100)
y = np.cos(x)
```

In [11]:
```
plt.plot(x,y)
```

Out[11]:
```
[<matplotlib.lines.Line2D at 0x7e6a2094f350>]
```



In [4]:
```
y
```

```
Out[4]: array([ 0.00000000e+00,  1.26592454e-01,  2.51147987e-01,  3.71662456e-01,
                4.86196736e-01,  5.92907929e-01,  6.90079011e-01,  7.76146464e-01,
                8.49725430e-01,  9.09631995e-01,  9.54902241e-01,  9.84807753e-01,
                9.98867339e-01,  9.96854776e-01,  9.78802446e-01,  9.45000819e-01,
                8.95993774e-01,  8.32569855e-01,  7.55749574e-01,  6.66769001e-01,
                5.67059864e-01,  4.58226522e-01,  3.42020143e-01,  2.20310533e-01,
                9.50560433e-02, -3.17279335e-02, -1.58001396e-01, -2.81732557e-01,
               -4.00930535e-01, -5.13677392e-01, -6.18158986e-01, -7.12694171e-01,
               -7.95761841e-01, -8.66025404e-01, -9.22354294e-01, -9.63842159e-01,
               -9.89821442e-01, -9.99874128e-01, -9.93838464e-01, -9.71811568e-01,
               -9.34147860e-01, -8.81453363e-01, -8.14575952e-01, -7.34591709e-01,
               -6.42787610e-01, -5.40640817e-01, -4.29794912e-01, -3.12033446e-01,
               -1.89251244e-01, -6.34239197e-02,  6.34239197e-02,  1.89251244e-01,
                3.12033446e-01,  4.29794912e-01,  5.40640817e-01,  6.42787610e-01,
                7.34591709e-01,  8.14575952e-01,  8.81453363e-01,  9.34147860e-01,
                9.71811568e-01,  9.93838464e-01,  9.99874128e-01,  9.89821442e-01,
                9.63842159e-01,  9.22354294e-01,  8.66025404e-01,  7.95761841e-01,
                7.12694171e-01,  6.18158986e-01,  5.13677392e-01,  4.00930535e-01,
                2.81732557e-01,  1.58001396e-01,  3.17279335e-02, -9.50560433e-02,
               -2.20310533e-01, -3.42020143e-01, -4.58226522e-01, -5.67059864e-01,
               -6.66769001e-01, -7.55749574e-01, -8.32569855e-01, -8.95993774e-01,
               -9.45000819e-01, -9.78802446e-01, -9.96854776e-01, -9.98867339e-01,
               -9.84807753e-01, -9.54902241e-01, -9.09631995e-01, -8.49725430e-01,
               -7.76146464e-01, -6.90079011e-01, -5.92907929e-01, -4.86196736e-01,
               -3.71662456e-01, -2.51147987e-01, -1.26592454e-01, -4.89858720e-16])
```

```
In [5]: x[0]
```

```
Out[5]: 0.0
```

```
In [6]: x[1]
```

```
Out[6]: 0.12693303650867852
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]: def integrate(function, x0, xf, steps):
            dx = (xf-x0)/steps
            for i in
```

```
In [ ]:
```

```
In [ ]:
```

In [ ]:

In [ ]:

In [ ]:

```
function(x0)
```